Automated Land-Cover Classification via Spectral Band Ratioing and Spatial Texture Analysis for Defense GIS

Shreyans Lunkad [BT24ECI042], Vasu Agrawal [BT24ECI008],
Naresh Bilavath [BT24ECI004] Tanush Cherukuri [BT24ECI007]

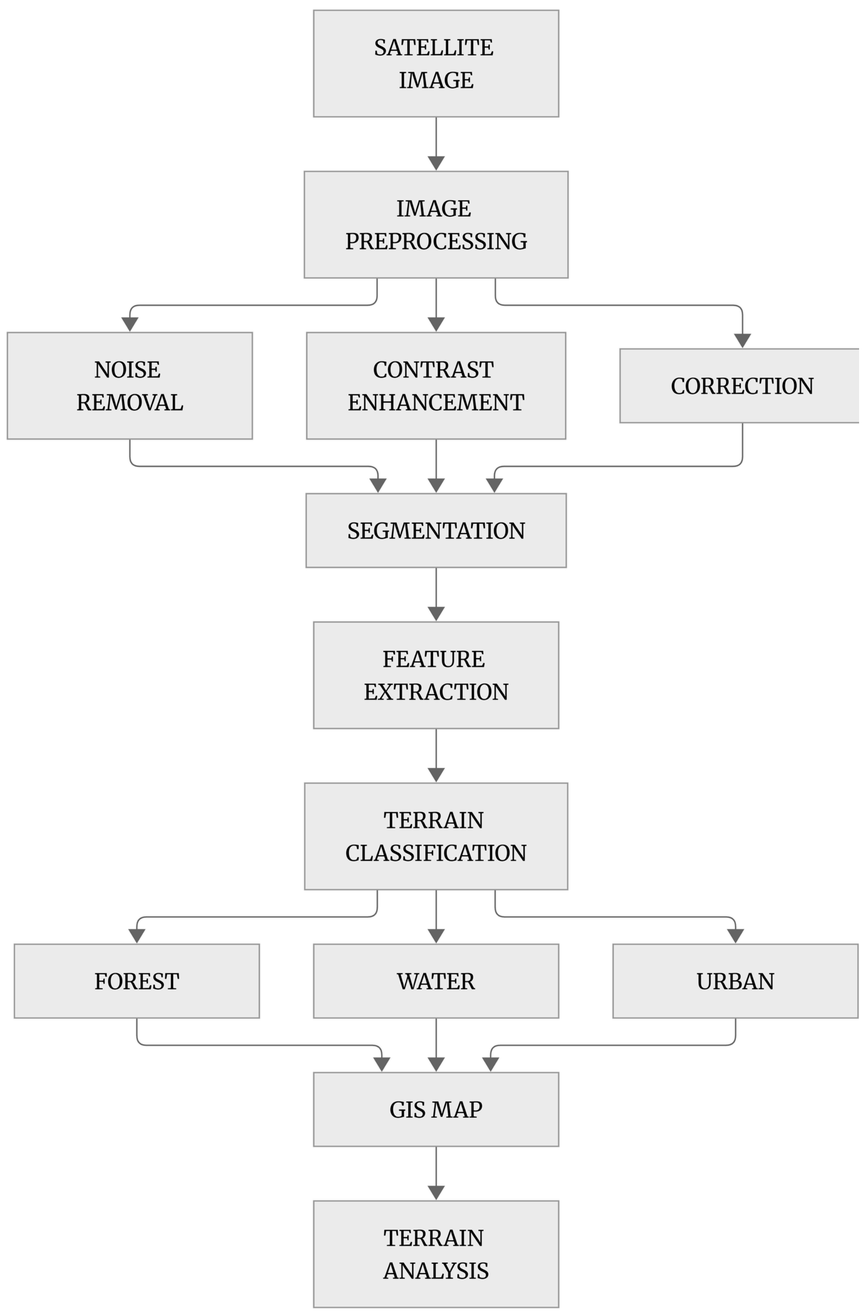
Indian Institute of Information Technology, Nagpur

Abstract

- This project explores the integration of Digital Image Processing (DIP) and Geographic Information Systems (GIS) for satellite-based terrain mapping and analysis.
- Image preprocessing, noise removal, contrast enhancement, segmentation, and feature extraction are applied to improve image quality and identify geographical features.
- Terrain classification techniques are used to categorize regions into forests, water bodies, roads, urban areas, agricultural land, and barren land.
- The classified output is integrated into GIS software to generate colour-coded thematic maps for clear visualization and spatial analysis.
- This approach can support military terrain awareness, border-area monitoring, environmental surveillance, and disaster management by converting satellite images into meaningful geographical information.

Introduction

- Satellite imagery plays an important role in understanding geographical landscapes and identifying different terrain features over large areas.
- Digital Image Processing techniques enhance satellite images by reducing noise, improving contrast, and highlighting important features for accurate analysis.
- Image segmentation and classification help distinguish terrain categories such as vegetation, water bodies, roads, settlements, and open land.
- Geographic Information Systems (GIS) combine classified image data with geographical coordinates to create layered maps that can be analyzed efficiently.
- The integration of DIP and GIS provides a systematic approach to terrain mapping and spatial analysis, with applications in military terrain assessment, infrastructure planning, environmental monitoring, and emergency response.
- The main objective is to demonstrate how satellite image processing and GIS can transform raw imagery into useful terrain information for improved geographical understanding and decision-making.



Flow - Chart for the Complete Process

Methodology

- GIS and Image Processing for Military Terrain Mapping
- **Satellite Image Acquisition**: Obtain satellite images of the selected geographical region from publicly available sources for terrain analysis.
- **Image Preprocessing**:The input image is preprocessed using noise reduction and image enhancement techniques to improve visual quality and preserve important terrain details.
- **Image Enhancement**: Apply contrast stretching and histogram equalization to improve the visibility of terrain features.
- **Image Segmentation**: Divide the satellite image into meaningful regions using techniques such as K-means clustering and thresholding.
- **Feature Extraction**: Extract colour, texture, spectral, and spatial features to distinguish different terrain types.
- **Terrain Classification**: Classify image regions into categories such as forests, water bodies, roads, urban areas, agricultural land, and bare land.
- **GIS Integration**: Import the classified results into GIS software and represent each terrain category as a separate geographical layer.
- **Map Visualization and Analysis**: Generate a colour-coded terrain map to visualize geographical features and support terrain assessment and environmental monitoring.
- **Result Validation**: Compare the classified map with reference data or manually labelled samples to evaluate classification accuracy.

Experiments and Results

Experiment	Technique	Observed result
Noise reduction	Gaussian/median filtering	Reduced small-scale image variations
Contrast enhancement	CLAHE	Improved local contrast and terrain visibility
Histogram analysis	Intensity distribution	Shows changes in pixel intensity distribution
Image segmentation	K-means clustering (\(K=5\))	Groups pixels into five color-based clusters
Edge detection	Canny operator	Highlights prominent boundaries and ridges
GIS integration	QGIS	Enables geographic visualization when data is georeferenced

Key Findings

- Histogram comparison provides a visual assessment of intensity redistribution.
- K-means segmentation separates pixels into color-based groups.
- Canny edge detection reveals boundaries and terrain structures.
- GIS integration provides a geographic framework for displaying and interpreting processed imagery.

Limitations

- Color-based clusters can combine different terrain types with similar colors.
- Snow, shadows, and rocky surfaces may be difficult to distinguish reliably.
- Accurate GIS overlays require suitable georeferencing.
- Quantitative classification accuracy requires reference labels or ground-truth data.

Important Point

Implementation Details

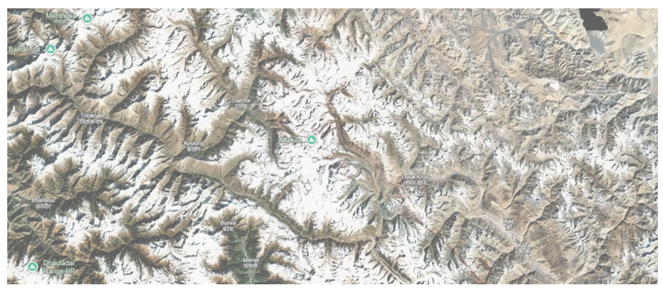


Fig.A : Original Input Image



Fig.B : Enhanced Input Image

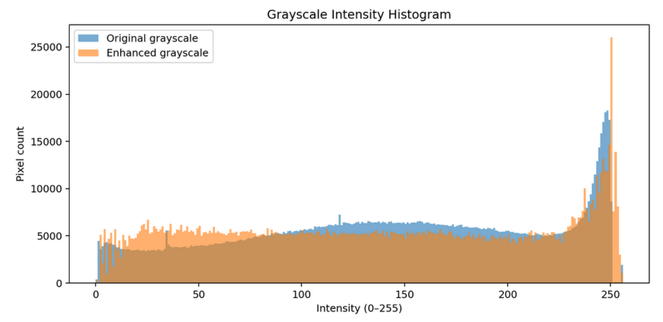


Fig.C:Histogram Comparison of the Original and Enhanced Images

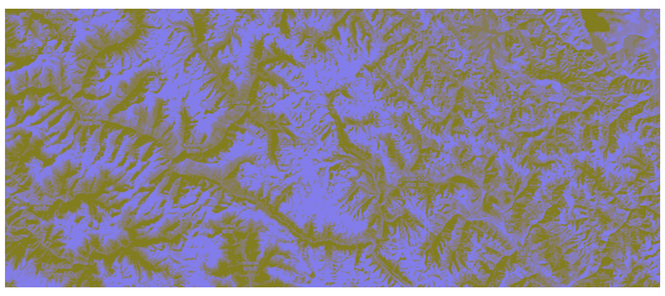


Fig.D :K-means clusterred image

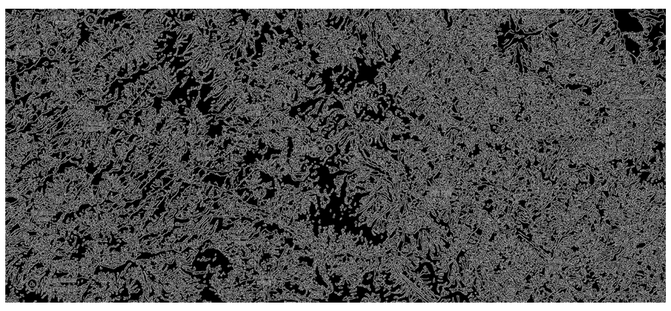


Fig.E :Edge Detection

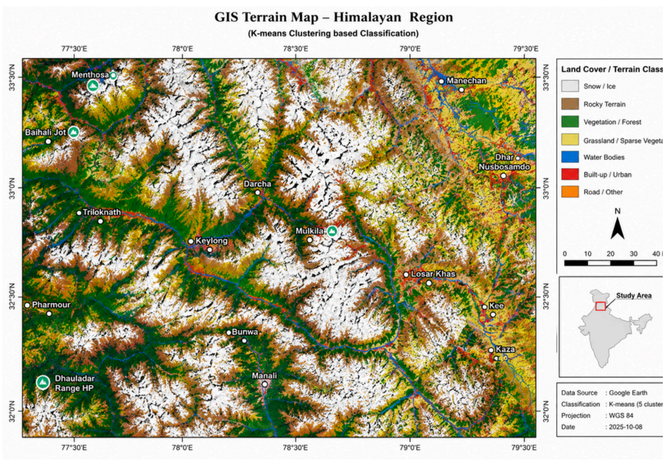


Fig.E : GIS Final Output

Conclusion

- DIP techniques enhance satellite image contrast and improve the visibility of terrain features for better analysis.
- K-means clustering and edge detection help identify color-based regions and highlight terrain boundaries.
- GIS integration enables geographic visualization and supports terrain mapping and spatial interpretation.

Acknowledgments

We sincerely thank our mentor, Dr. Tapan Jain, and all faculty members for their invaluable guidance, support, and encouragement through this project. Their insights played a crucial role in shaping this work.

References

[1]. Canny, J. (1986). “A Computational Approach to Edge Detection.” IEEE Transactions on Pattern Analysis and Machine Intelligence, 8(6), 679–698.
[2]. “Improving K-Means Clustering: A Comparative Study of Parallelized Version of Modified K-Means Algorithm for Clustering of Satellite Images.” (2025). Algorithms, 18(8), 532.